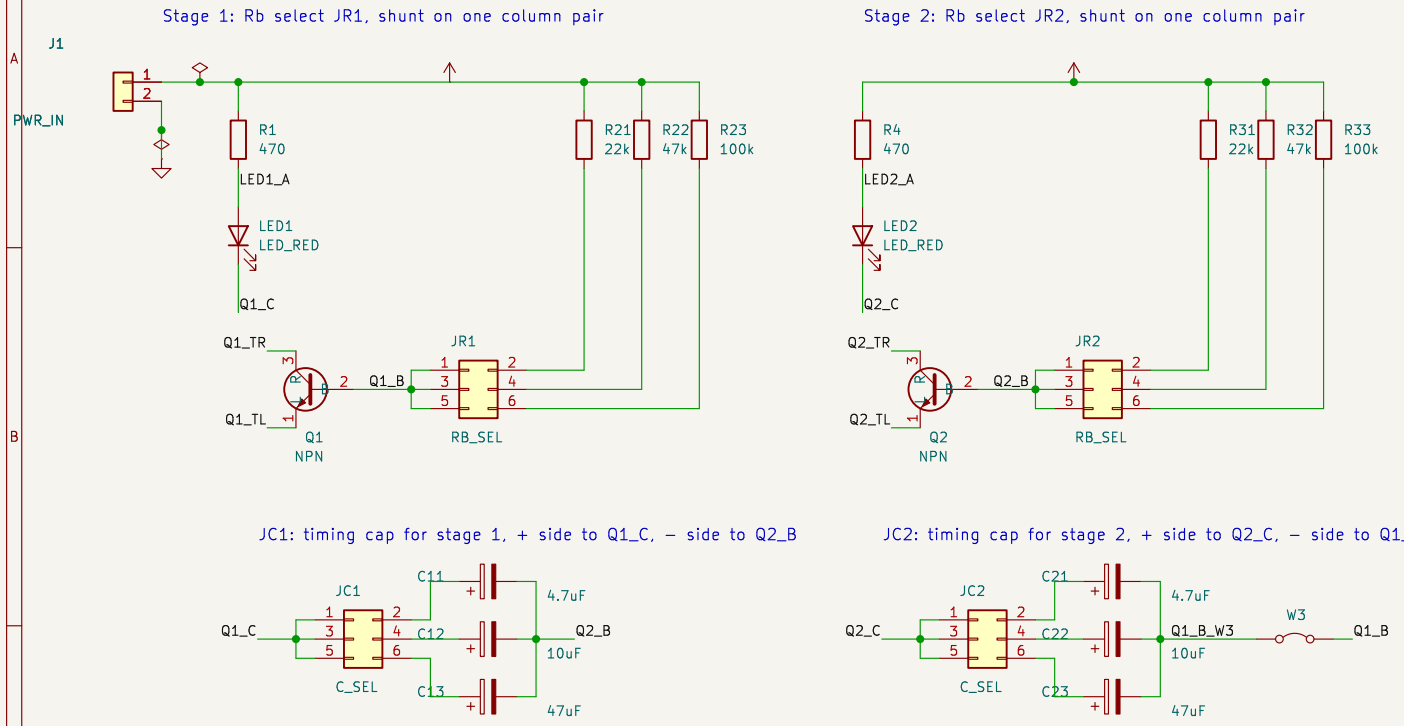
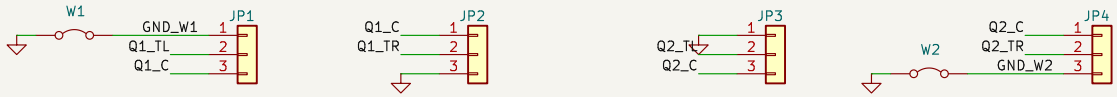


Pinout select, per transistor: pad 1 = GND (emitter), pad 2 = TO−92 outer leg, pad 3 = collector.
Shunt 1−2 on both = E−B−C (2N3904, PN2222, S8050). Shunt 2−3 on both = C−B−E (BC547, BC548, BC337).



W1 / W2 / W3: bare wire links on the component side (single−sided board); each joins two copper islands of one net.

Astable multivibrator LED flasher, v2: single copper layer (bottom), home etchable, all through−hole.
JR1 / JR2 pick the base resistor (22k / 47k / 100k), JC1 / JC2 pick the timing cap (4.7u / 10u / 47u): one shunt per header.
f = 1 / (1.386 * Rb * C): 22k/4.7u 7 Hz ... 47k/10u 1.5 Hz ... 100k/47u 0.15 Hz. Supply 3 to 9 V, 5 V recommended.
Unselected caps float on the + side and stay tied to the base on the − side; unselected resistors float at the base end.

Home etchable: bottom copper only, 1.0 mm tracks, 0.6 mm clearance, three wire links
MIT licensed hobby project

Sheet: /
File: astable_v2.kicad_sch

Title: Astable multivibrator LED flasher v2 − single sided, selectable RC

Size: A4	Date: 2026−10−21	**Rev: 2.0**
KiCad E.D.A. 9.0.8		Id: 1/1